

Q1: 24 Feb (Shift 1) - Single Correct

A scientific committee is to be formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways, the committee can be formed is:

- (1) 560
- (2) 1050
- (3) 1625
- (4) 575

Q2: 24 Feb (Shift 2) - Numerical

The students $S_1, S_2, \dots, S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is

Q3: 25 Feb (Shift 1) - Single Correct

The total number of positive integral solutions (x, y, z) such that $xyz = 24$ is

- (1) 36
- (2) 45
- (3) 24
- (4) 30

Q4: 25 Feb (Shift 1) - Numerical

The total number of numbers, lying between 100 and 1000 that can be formed with the digits 1, 2, 3, 4, 5, if the repetition of digits is not allowed and numbers are divisible by either 3 or 5 is

Q5: 25 Feb (Shift 2) - Numerical

The total number of two digit numbers 'n', such that $3^n + 7^n$ is a multiple of 10, is

Q6: 26 Feb (Shift 1) - Single Correct

The number of seven digit integers with sum of the digits equal to 10 and formed by using the digits 1,2 and 3 only is

(1) 77

(2) 42

(3) 35

(4) 82

Q7: 26 Feb (Shift 2) - Single Correct

A natural number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}$, $y > z$. Then the number of odd divisors of n , including 1, is :

(1) 11

(2) $6x$

(3) 12

(4) 6

Q8: 26 Feb (Shift 2) - Numerical

The total number of 4 -digit numbers whose greatest common divisor with 18 is 3 , is

Answer Key**Q1 (3)****Q2 (31650)****Q3 (4)****Q4 (32)****Q5 (45)****Q6 (1)****Q7 (3)****Q8 (1000)**

Q1 (3)

$$(2I, 4 F) + (3I, 6 F) + (4I, 8 F)$$

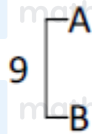
$$= {}^6C_2 {}^8C_4 + {}^6C_3 {}^8C_6 + {}^6C_4 {}^8C_8$$

$$= 15 \times 70 + 20 \times 28 + 15 \times 1$$

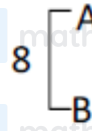
$$= 1050 + 560 + 15 = 1625$$

Q2 (31650)

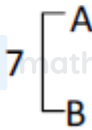
C → 1



C → 2



C → 3



$$= {}^{10}C_1 [2^9 - 2] + {}^{10}C_2 [2^8 - 2] + {}^{10}C_3 [2^7 - 2]$$

$$= 2^7 [{}^{10}C_1 \times 4 + {}^{10}C_2 \times 2 + {}^{10}C_3] - 20 - 90 - 240$$

$$= 128[40 + 90 + 120] - 350$$

$$= (128 \times 250) - 350$$

$$= 10[3165] = 31650$$

Q3 (4)

$$x \cdot y \cdot z = 24$$

$$x \cdot y \cdot z = 2^3 \cdot 3^1$$

Now using beggars method. 3 things to be distributed among 3 persons Each may receive none, one or more

$$\therefore {}^5C_2 \text{ ways}$$

Hints and Solutions

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Similarly for '1' $\therefore {}^3C_2$ ways

Total ways $= {}^5C_2 \cdot {}^3C_2 = 30$ ways

Q4 (32)

$\square\square\square$

divisible by $\rightarrow 3$

divisible by 5

$\square\square\square$

$= 12$

4×3

$12 \rightarrow 3, 4, 5 \rightarrow 3! = 6$

$15 \rightarrow 2, 3, 4 \rightarrow 3! = 6$

$24 \rightarrow 1, 3, 5 \rightarrow 3! = 6$

$42 \rightarrow 1, 2, 3 \rightarrow 3! = 6$

Required No. $= 24 + 12 - 4 = 32$

Q5 (45)

$$\therefore 7^n = (10 - 3)^n = 10K + (-3)^n$$

$$\therefore 7^n + 3^n = 10K + (-3)^n + 3^n$$

$\rightarrow 10K$ if $n = \text{odd}$

$\rightarrow 10K + 2 \cdot 3^n$ if $n = \text{even}$

Let $n = 2t; t \in \mathbb{N}$

$$\therefore 3^n = 3^{2t} = (10 - 1)^t$$

$$= 10p + (-1)^t$$

$$= 10p \pm 1$$

$\therefore$ if $n = \text{even}$ then $7^n + 3^n$ will not be multiply of 10

So if n is odd then only $7^n + 3^n$ will be multiply of 10

$$\therefore n = 11, 13, 15, \dots, 99$$

$\therefore$ Ans 45

Q6 (1)

CASE-I: 1,1,1,1,1,2,3

$$\text{WAYS} = \frac{7!}{5!} = 42$$

CASE-II: 1,1,1,1,2,2,2

$$\text{WAYS} = \frac{7!}{4! \cdot 3!} = 35$$

$$\text{TOTAL WAYS} = 42 + 35 = 77$$

Q7 (3)

$$y + z = 5 \dots (1)$$

$$\frac{1}{y} + \frac{1}{z} = \frac{5}{6}$$

$$\Rightarrow \frac{y+z}{yz} = \frac{5}{6}$$

$$\Rightarrow \frac{5}{yz} = \frac{5}{6}$$

$$\Rightarrow yz = 6$$

$$\text{Also } (y - z)^2 = (y + z)^2 - 4yz$$

$$\Rightarrow (y - z)^2 = (y + z)^2 - 4yz$$

$$\Rightarrow (y - z)^2 = 25 - 4(6) = 1$$

$$\Rightarrow y - z = 1$$

from (1) and (2), $y = 3$ and $z = 2$

for calculating odd divisor of $p = 2^x \cdot 3^y \cdot 5^z$

x must be zero

$$P = 2^0 \cdot 3^3 \cdot 5^2$$

$$\therefore \text{total odd divisors must be } (3 + 1)(2 + 1) = 12$$

Q8 (1000)

Since, required number has G.C.D with 18 as 3. It must be odd multiple of '3' but not a multiple of '9'.

(i) Now, 4 -digit number which are odd multiple of '3' are,

$$1005, 1011, 1017, \dots, 9999 \rightarrow 1499$$

Hints and Solutions

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(ii) 4-digit number which are odd multiple of 9 are,

1017, 1035, 9999 $\rightarrow$ 499

$\therefore$ Required numbers = $1499 - 499 = 1000$